

Bidyut Bikash Goswami

Department of Mathematics
University of Exeter
North Park Road
Exeter EX4 4QE, UK

B.Goswami2@exeter.ac.uk
<https://bidyutbg.github.io/>
+44 7350 184177

RESEARCH INTEREST

The Indian summer monsoon is a central component of my research, which focuses on understanding atmospheric dynamics and variability, and their response to warming, with particular emphasis on atmospheric convection, its organisation, and the resulting dynamical interactions.

EDUCATION

PhD Atmospheric and Space Sciences, Indian Institute of Tropical Meteorology, Pune, India (2014)
MSc Physics, Gauhati University, Assam, India (2006)

EMPLOYMENT PROFILE

2025- University of Exeter, UK, with Robin Chadwick
2022-2025 Inst. of Science and Technology, Austria, with Caroline Muller
2019-2022 Yonsei University, South Korea, with Soon-Il An
2017-2019 New York University Abu Dhabi, UAE, with Andrew Majda
2014-2017 University of Victoria, Canada, with Boualem Khouider
2009-2014 Indian Institute of Tropical Meteorology, India, with P. Mukhopadhyay and B.N. Goswami

GRANTS AWARDED

2025–2027 “A Data-Driven Multicloud and Multiscale Parameterization of Organized Convection to Improve the Predictions of Monsoon Extreme Weather and Extreme Precipitation Events.”
Co-Principal Investigator with PI: Boualem Khouider (UVic) and Co-PI: Courtney Schumacher (TAMU). USD 450,000. Monsoon Mission, Ministry of Earth Sciences, India

TEACHING & MENTORING

2022–2024	Weather and Climate: Apps to Maps	Graduate course at ISTA
2022–2024	Climate Modeling	Graduate course at ISTA
2024	Tropical Meteorology	Graduate course, University of Vienna

2022–2024 2 Master’s theses, 1 pre-doctoral internship, 4 ISTA Rotation projects
(co-Advising with Caroline Muller)

AWARDS & HONORS

2022 Professor Roddam Narasimha Best Research Award, WASAMSA22
2013 Best Presentation Award, Indian Meteorological Society Pune Chapter
2012 International Travel Support, Science and Engineering Research Board, India
2011 Best paper (Silver Jubilee award), Indian Institute of Tropical Meteorology

PUBLICATIONS

Journal Articles

- [1] B. B. Goswami, Z. Lu, and C. Muller. “Convective self-aggregation in diurnally oscillating sea surface temperature and solar forcing experiments”. In: *Journal of Advances in Modeling Earth Systems* 18.1 (2026), e2024MS004576.
- [2] B. B. Goswami and C. J. Muller. “What advances monsoon onset over India?” In: *Weather and Climate Dynamics* 7.2 (2026), pp. 583–596.
- [3] B. B. Goswami, A. Polesello, and C. Muller. “An assessment of representing land-ocean heterogeneity via CAPE relaxation timescale in the Community Atmospheric Model 6 (CAM6)”. In: *Journal of Advances in Modeling Earth Systems* 17.9 (2025), e2025MS005035.
- [4] B. B. Goswami. “A pre-monsoon signal of false alarms of Indian monsoon droughts”. In: *Geophysical Research Letters* 51.5 (2024), e2023GL106569.
- [5] S. Paik, D. Kim, S.-I. An, H. Oh, J. Shin, B. B. Goswami, S.-K. Min, and S. K. Mondal. “Exploring causes of distinct regional and subseasonal Indian summer monsoon precipitation responses to CO2 removal”. In: *npj Climate and Atmospheric Science* 7.1 (2024), p. 305.
- [6] B. B. Goswami. “Role of the eastern equatorial Indian Ocean warming in the Indian summer monsoon rainfall trend”. In: *Climate Dynamics* 60.1 (2023), pp. 427–442.
- [7] B. B. Goswami and S.-I. An. “An assessment of the ENSO-monsoon teleconnection in a warming climate”. In: *npj Climate and Atmospheric Science* 6.1 (2023), p. 82.
- [8] B. Khouider, B. B. Goswami, R. Phani, and A. Majda. “A Shallow-Deep Unified Stochastic Mass Flux Cumulus Parameterization in the Single Column Community Climate Model”. In: *Journal of Advances in Modeling Earth Systems* 15.11 (2023).
- [9] B. B. Goswami, S.-I. An, and R. Murtugudde. “Role of the Tibetan plateau glaciers in the Asian summer monsoon”. In: *Climatic Change* 173.3 (2022), p. 29.
- [10] B. B. Goswami, R. Murtugudde, and S.-I. An. “Role of the Bay of Bengal warming in the Indian summer monsoon rainfall trend”. In: *Climate Dynamics* 59.5 (2022), pp. 1733–1751.
- [11] K. Roy, P. Mukhopadhyay, R. Krishna, B. Khouider, and B. B. Goswami. “Evaluation of mean state in NCEP Climate Forecast System (version 2) simulation using a stochastic multicloud model calibrated with DYNAMO RADAR data”. In: *Earth and Space Science* 8.8 (2021), e2020EA001455.
- [12] T. Goswami, B. B. Goswami, R. P. M. Krishna, and P. Mukhopadhyay. “Evaluation of SP-CAM and SP-CCSM in capturing the extremes of summer monsoon rainfall over Indian region”. In: *Journal of Earth System Science* 129.1 (2020), p. 116.
- [13] B. B. Goswami, B. Khouider, R. Phani, P. Mukhopadhyay, and A. Majda. “The Stochastic Multi-cloud Model (SMCM) Convective Parameterization in the CFSv2: Scopes and Opportunities”. In: *Current Trends in the Representation of Physical Processes in Weather and Climate Models* (2019), pp. 157–181.
- [14] B. B. Goswami, B. Khouider, R. Phani, P. Mukhopadhyay, and A. J. Majda. “Implementation and calibration of a stochastic multicloud convective parameterization in the NCEP Climate Forecast System (CFSv2)”. In: *Journal of Advances in Modeling Earth Systems* (2017).
- [15] B. B. Goswami, B. Khouider, R. Phani, P. Mukhopadhyay, and A. Majda. “Improved tropical modes of variability in the NCEP Climate Forecast System (version 2) via a stochastic multicloud model”. In: *Journal of the Atmospheric Sciences* 74.10 (2017), pp. 3339–3366.
- [16] B. B. Goswami and B. N. Goswami. “A road map for improving dry-bias in simulating the South Asian monsoon precipitation by climate models”. In: *Climate Dynamics* (2016).
- [17] B. B. Goswami, B. Khouider, R. P. M. Krishna, P. Mukhopadhyay, and A. J. Majda. “Improving Synoptic and Intra-Seasonal Variability in CFSv2 via Stochastic Representation of Organized Convection”. In: *Geophysical Research Letters* (2016).

- [18] B. B. Goswami, R. Krishna, P. Mukhopadhyay, M. Khairoutdinov, and B. Goswami. “Simulation of the Indian summer monsoon in the superparameterized Climate Forecast System version 2: Preliminary results”. In: *Journal of Climate* 28.22 (2015), pp. 8988–9012.
- [19] B. B. Goswami, M. Deshpande, P. Mukhopadhyay, S. K. Saha, S. A. Rao, R. Murthugudde, and B. Goswami. “Simulation of monsoon intraseasonal variability in NCEP CFSv2 and its role on systematic bias”. In: *Climate dynamics* 43.9 (2014), pp. 2725–2745.
- [20] B. B. Goswami, P. Mukhopadhyay, M. Khairoutdinov, and B. Goswami. “Simulation of Indian summer monsoon intraseasonal oscillations in a superparameterized coupled climate model: Need to improve the embedded cloud resolving model”. In: *Climate dynamics* 41.5 (2013), pp. 1497–1507.
- [21] B. B. Goswami, N. J. Mani, P. Mukhopadhyay, D. E. Waliser, J. J. Benedict, E. D. Maloney, M. Khairoutdinov, and B. Goswami. “Monsoon intraseasonal oscillations as simulated by the superparameterized Community Atmosphere Model”. In: *Journal of Geophysical Research: Atmospheres* 116.D22 (2011).
- [22] B. B. Goswami, P. Mukhopadhyay, R. Mahanta, and B. Goswami. “Multiscale interaction with topography and extreme rainfall events in the northeast Indian region”. In: *Journal of Geophysical Research: Atmospheres* 115.D12 (2010).

Conference Papers

- [1] P. Mukhopadhyay, R. P. M. Krishna, B. B. Goswami, S. Abhik, M. Ganai, M. Mahakur, M. Khairoutdinov, and J. Dudhia. “Improvement of Systematic Bias of mean state and the intraseasonal variability of CFSv2 through superparameterization and revised cloud-convection-radiation parameterization”. In: *Remote Sensing and Modeling of the Atmosphere, Oceans, and Interactions VI*. Vol. 9882. SPIE. 2016, pp. 118–123.

Book Chapters

- [1] B. Khouider, B. B. Goswami, and P. M. Krishna. “Toward a breakthrough in convection parameterization using multiscale and multicloud modeling constrained by observations—Part I: Motivation and model development”. In: *Multi-Scale Precipitation Variability Over the Tropics*. Elsevier, 2025, pp. 105–140.
- [2] B. Khouider, K. Roy, B. B. Goswami, and P. M. Krishna. “Toward a breakthrough in convection parameterization using multiscale and multicloud modeling constrained by observations—Part II: Implementation in CAM and preliminary results”. In: *Multi-Scale Precipitation Variability Over the Tropics*. Elsevier, 2025, pp. 105–140.
- [3] B. B. Goswami, B. Khouider, R. Phani, P. Mukhopadhyay, and A. Majda. “The Stochastic Multi-cloud Model (SMCM) Convective Parameterization in the CFSv2: Scopes and Opportunities”. In: *Current Trends in the Representation of Physical Processes in Weather and Climate Models*. Springer, 2019, pp. 157–181.

Technical Reports

- [1] P. M. R. Krishna, S. Abhik, B. B. Goswami, M. Ganai, M. Mahakur, M. Deshpande, P. Mukhopadhyay, R. S. Das, M. Khairoutdinov, J. Dudhia, and B. N. Goswami. *Improvement of Cloud and Convective Parameterizations in Climate Forecast System Version 2 (CFSv2): Indian Summer Monsoon Perspective*. IITM Research Report Research Report No. RR-135. ESSO/IITM/SERP/SR/01(2016)/186. Indian Institute of Tropical Meteorology (IITM), Earth System Science Organisation (ESSO), 2016.

CONFERENCE PRESENTATIONS

Invited Talks

- [1] *False alarms of Indian monsoon droughts*. AGU Atmospheric Science Early Career Webinar. Invited talk, August 13, 2024. Aug. 2024.
- [2] *Pre-monsoon rainfall over northeast India: A signal of false alarms of Indian monsoon droughts*. 58th Prof. R. Ananthakrishnan Colloquium, Ministry of Earth Sciences. Invited talk. India, Apr. 2024.
- [3] *Indian Ocean warming and its impact on the Indian summer monsoon*. Virtual International Conference on Impacts & Consequences of Environmental Degradation on Animal Health and Human Wellbeing. Invited talk, Abhayapuri College and Gauhati University. Sept. 2021.
- [4] *Stochastic Multi-cloud Model (SMCM) in the Climate Forecast System Model (version 2)*. Workshop on Moist Processes in the Atmosphere. Invited talk. Mathematisches Forschungsinstitut Oberwolfach, Germany, 2019.
- [5] *An attempt to break the deadlock: A stochastic convective parameterization approach*. Indian Institute of Tropical Meteorology. Invited seminar. Pune, India, 2017.
- [6] *Better representation of organized convection in the Climate Forecast System via Stochastic Multi-Cloud Model*. Center for Ocean-Land-Atmosphere Studies, George Mason University. Invited talk. Fairfax, USA, 2016.
- [7] *Is Superparameterization capable of breaking the “deadlock”? Seeking the answer in Superparameterized CFSv2 664-day climate*. Workshop on Stochasticity and Organization of Tropical Convection. Invited talk. Banff International Research Station, Canada, 2015.
- [8] *Multiscale interaction with topography and extreme rainfall events in the North-East Indian region*. 23rd IITM Silver Jubilee Award Lecture. Invited lecture. Pune, India, 2011.

Contributed Talks

- [1] *What advances monsoon onset over India?* European Geosciences Union General Assembly. Vienna, Austria, 2024.
- [2] *A shallow-deep unified stochastic mass flux cumulus parameterization*. Workshop on Atmospheric Satellites for Asian Monsoon Cloud-Precipitation Science. Bengaluru, India, 2022.
- [3] *Stochastic plume ensembles for a unified shallow-deep mass flux cumulus parameterization in CESM*. Convection Parametrization: Progress and Challenges Workshop (CPPC2019). Met Office, Exeter, UK, 2019.
- [4] *Accurate representation of organized convection in CFSv2 via a stochastic lattice model*. 97th American Meteorological Society Annual Meeting. Seattle, USA, 2017.
- [5] *Superparameterized Climate Forecast System for improved simulation of Indian summer monsoon*. World Weather Open Science Conference. Montreal, Canada, 2014.
- [6] *Superparameterization – a promise towards better GCM simulation of Indian summer monsoon*. Annual Monsoon Workshop. Indian Institute of Tropical Meteorology, Pune, India, 2013.

Contributed Posters

- [1] *A pre-monsoon signal of false alarms of Indian monsoon droughts*. American Geophysical Union Annual Meeting. Washington DC, USA, 2024.
- [2] *How does a convection cluster respond to a large-scale mean wind?* American Geophysical Union Annual Meeting. Washington DC, USA, 2024.
- [3] *Response of convection organization to diurnally varying surface forcing*. Joint CFMIP-GASS Meeting. Sorbonne University, Paris, France, 2023.
- [4] *Response of the Indian monsoon to a warming Indian Ocean*. European Geosciences Union General Assembly. Vienna, Austria, 2023.

- [5] *A shallow-deep unified stochastic mass flux cumulus parameterization in CESM*. American Geophysical Union Fall Meeting. San Francisco, USA, 2019.
- [6] *Stochastic Multicloud Model implemented in CFSv2: Sensitivity to middle tropospheric moisture*. American Meteorological Society Annual Meeting. Austin, USA, 2018.
- [7] *A stochastic multicloud convective parameterization in the NCEP Climate Forecast System*. American Geophysical Union Fall Meeting. New Orleans, USA, 2017.
- [8] *Superparameterization: A promise towards a better Earth System Model*. International Conference on Earth System Modelling. Hamburg, Germany, 2012.

REVIEWER ACTIVITIES

Geophysical Research Letters, Journal of Climate, Climate Dynamics, International Journal of Climatology, Journal of the Atmospheric Sciences, Atmosphere, Climate, Water, Journal of Earth System Science, Pure and Applied Geophysics, Scientific Reports, Atmospheric Research, Quarterly Journal of the Royal Meteorological Society, Physics and Chemistry of the Earth, Journal of Atmospheric and Solar-Terrestrial Physics, Journal of Geophysical Research: Atmospheres, Geoscience Letters

OUTREACH ACTIVITIES

2022–present **Climate Dialogues: Way Forward**

It is an initiative series designed to spread awareness about Climate Science and excite young students to pursue careers in this field (especially designed for the people of Assam & other northeastern states of India). [Youtube Link: <https://t.ly/edZk8>]

Talks **Career in Earth Science**

Webinar by Advancing North East, an initiative by Ministry of Development of North Eastern Region (MDoNER) & North Eastern Council (NEC), implemented by North Eastern Development Finance Corporation Ltd. (NEDFi), India, June 5, 2022

Climate change Lessons and Actions

International Webinar organized by the Department of Zoology, Gauhati University, Zoological Society of Assam, and the Department of Zoology, Morigaon College, Assam, July 10, 2021

Atmospheric sciences, Climate change & our knowledge about it

‘Bigyan Sora’ by All India Radio Tezpur center, September 14, 2020

I shared my story

An interview by the AGU narratives, December 12, 2018

Other **Story of a Hailstone**

HUMAN-MODEL-WORLD: A collection of scientific images by BRIDGE Network, 2024